SFSR: Spectral Fusion Super-Resolution for Multisensor Remote Sensing With Degraded References

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***Abstract*—Reference-based super-resolution (RefSR) aims to enhance low-resolution (LR) imagery by leveraging auxiliary reference observations. While effective under controlled settings, most existing RefSR methods implicitly assume that reference images are clean or geometrically consistent with the target input. In real-world remote sensing systems, however, reference observations are frequently degraded by sensor noise, atmospheric blur, and geometric inconsistencies caused by different viewing angles and acquisition times. Moreover, due to the inherent resolution gap between LR and reference images, strict spectral consistency is difficult to guarantee in practice. These factors substantially reduce the reliability of reference cues and limit the applicability of RefSR in multisensor satellite imaging scenarios. To address these challenges, we propose spectral fusion super-resolution (SFSR), a diffusion-based RefSR framework designed to operate robustly under degraded reference conditions. At its core, SFSR introduces the spectral swin cross-attention module (S2CAM), which enables frequency-aware reference utilization and integrates the refined reference features as conditional guidance within the reverse diffusion process. By explicitly redistributing spectral components and suppressing unreliable high-frequency responses introduced by noise, SFSR enables stable and effective use of reference information that conventional RefSR methods struggle to exploit. Extensive experiments on synthetic and benchmark satellite datasets demonstrate that SFSR consistently outperforms state-of-the-art RefSR approaches in terms of peak signal-to-noise ratio (PSNR), similarity index measure (SSIM), and perceptual metrics while maintaining high visual fidelity under severe degradation. In addition, evaluations on downstream tasks such as object detection and semantic segmentation show that SFSR leads to clear performance improvements, confirming its robustness and practical value for real-world multisensor remote sensing applications.**

***Index Terms*—Diffusion probabilistic model (DPM), multisensor remote sensing, reference-based super-resolution (SR).**

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I. Introduction

SUPER-RESOLUTION (SR) in remote sensing has become an essential research topic, as spatial resolution plays a decisive role in applications such as Earth observation, urban planning, environmental monitoring, and disaster response [41], [42]. High-resolution (HR) satellite imagery enables more reliable identification of fine-scale features, including roads, buildings, and vegetation boundaries. However, the achievable spatial resolution of wide field-of-view (WFOV) imaging systems is often limited by hardware constraints, particularly those arising from compact payload design and wide-angle optics. These limitations typically lead to reduced effective resolution, optical blur, and oversmoothing, which in turn degrade the accuracy of downstream analyses.

A practical approach to mitigating this limitation is reference-based super-resolution (RefSR) [6], which enhances a low-resolution (LR) image by leveraging auxiliary reference images that contain complementary structural or textural information. In satellite imaging, telephoto narrow field-of-view (NFOV) sensors can serve as valuable reference sources for improving WFOV imagery, especially in multicamera or multisensor satellite systems. Such configurations allow wide-area coverage to be combined with localized high-detail observations without relying on costly global HR acquisitions. Despite its promise, applying RefSR to real satellite imagery remains challenging. Most existing RefSR models for remote sensing implicitly assume that reference images are clean and geometrically consistent with the target WFOV image [38]. In practical remote sensing scenarios, these assumptions rarely hold. NFOV reference images are frequently degraded by atmospheric scattering, optical blur, and sensor noise [39], and they may only partially overlap with the target image. Moreover, even after orthorectification, differences in acquisition geometry, viewing angle, and acquisition time across sensors often introduce residual subpixel misalignments and illumination inconsistencies [40]. Such preprocessing also incurs additional computational cost and operational overhead. These factors make reliable structural correspondence difficult

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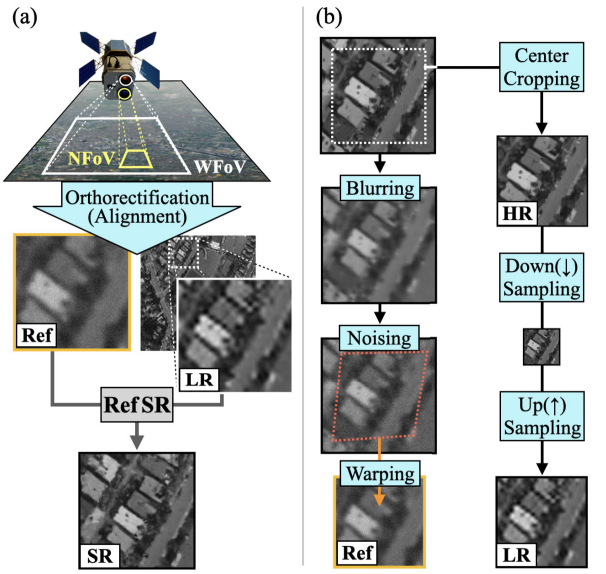


Fig. 1. Problem setup. (a) WFOV (LR) image with a degraded NFOV reference (Ref, yellow box) captured by a multicamera satellite system. (b) Synthetic degradation pipeline for generating WFOV (LR) and NFOV (Ref) image pairs under realistic multisensor imaging conditions.

and significantly reduce the robustness of conventional RefSR methods. Fig. 1(a) illustrates a representative example in which a WFOV LR image is paired with a degraded NFOV reference (Ref) captured by a multicamera satellite system.

Due to the lack of publicly available multiview satellite datasets that capture such realistic degradations, we construct a synthetic degradation pipeline. As illustrated in Fig. 1(b), this pipeline emulates practical WFOV and NFOV imaging conditions by incorporating physically motivated degradations, including optical blur, Gaussian noise, and random spatial perturbations. This enables controlled training and evaluation of RefSR models under realistic satellite imaging conditions and facilitates systematic analysis of how different degradations affect reconstruction accuracy.

Building upon this foundation, we propose spectral fusion super-resolution (SFSR), a diffusion-based RefSR framework designed to address the inherent resolution imbalance and reference unreliability between NFOV and WFOV satellite imagery arising from their distinct acquisition characteristics. SFSR aims to integrate the reliable global structural information provided by WFOV images with the complementary fine details present in NFOV references in a controlled and robust manner.

At the core of SFSR is the spectral swin cross-attention module (S2CAM), which enables effective reference utilization through spectral decomposition and noise-aware feature handling. S2CAM consists of two key submodules: the spectral splitter (SSP), which decomposes reference features into frequency-specific components to construct representations better suited for cross-attention-based fusion, and the noise splitter (NSP), which separates noise-dominated regions from the high-frequency reference features while preserving edge-relevant structural information. Together, these components allow SFSR to exploit informative reference cues while preventing noise-contaminated responses from interfering with the reconstruction process under degraded conditions.

In addition, two frequency-aware loss functions are introduced to guide effective reference utilization during training. The low-frequency alignment loss (LF Loss) enforces consistency between the low-frequency components of the WFOV input and the reference features, enabling reliable use of complementary higher frequency information from the reference. The noise minimization loss (NM Loss) suppresses noise-dominated high-frequency responses in degraded NFOV features, reducing the influence of unreliable reference signals. Together, these losses support stable spectral redistribution during the reverse diffusion process, allowing SFSR to progressively reconstruct HR satellite images with improved spatial precision and structural fidelity.

Our main contributions are summarized as follows.

1) We develop a synthetic degradation pipeline that simulates realistic WFOV/NFOV satellite imaging conditions, enabling controlled evaluation of RefSR under noise, blur, and geometric misalignment.
2) We propose SFSR, a diffusion-based RefSR framework that addresses resolution imbalance and reference unreliability between WFOV and NFOV images through frequency-aware reference utilization.
3) We demonstrate the effectiveness of SFSR through extensive experiments on multicamera satellite datasets, showing consistent improvements over existing RefSR methods in reconstruction quality and downstream tasks, including object detection and small-object segmentation.

II. Related Works

A. Super-Resolution in Remote Sensing

Super-resolution (SR) is a fundamental problem in remote sensing, as improved spatial resolution directly benefits downstream tasks such as object detection and land-cover analysis. Early model-based approaches [48], [49], [50] relied on handcrafted priors and iterative optimization, but often produced oversmoothed results and exhibited limited robustness across different sensors and imaging conditions.

Deep learning has significantly advanced remote sensing SR. Convolutional models such as SRCNN [43] and EDSR [44] enabled end-to-end LR–HR mapping, while transformer-based methods like SwinIR [12] further improved performance by modeling long-range dependencies and global context in large-scale satellite imagery.

More recently, diffusion-based models have been explored for remote sensing SR due to their strong generative capability and robustness to complex degradations. EDiffSR [47] introduced diffusion probabilistic models for remote sensing imagery, achieving improved perceptual quality while maintaining training stability through efficient noise prediction and enhanced conditional priors. RefDiff [38] further extended diffusion-based SR to reference-guided settings by incorporating land-cover change priors, demonstrating strong perceptual performance, particularly in terms of FID.

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Despite these advances, diffusion-based approaches still face challenges in preserving structural fidelity under large scaling factors, especially when reference information is degraded or affected by geometric distortion. These limitations become more pronounced in realistic multisensor satellite scenarios, motivating the need for more controlled and interpretable mechanisms for utilizing reference information.

B. Reference-Based Super-Resolution

RefSR aims to overcome the limitations of single-image SR (SISR) by exploiting external reference images that provide complementary structural or textural information. Early methods such as SRNTT [6] relied on patch matching and explicit feature alignment, but were highly sensitive to geometric misalignment and reference degradation. Subsequent approaches, including DCSR [7] and MASA [9], introduced deformable or coarse-to-fine alignment strategies to improve robustness under moderate misalignment.

Transformer-based models such as TTSR [8] and DATSR [10] reformulated RefSR as a cross-attention problem, enabling more flexible feature interaction and multiscale fusion. KeDuSR [36] further explored dual-lens configurations by explicitly modeling the resolution gap between wide-angle and telephoto sensors in cross-sensor RefSR scenarios.

Diffusion-based RefSR methods have also attracted growing attention. RefDiff [38], in particular, demonstrates strong perceptual performance and competitive FID scores, indicating that diffusion models can synthesize visually plausible HR details from reference guidance. However, these methods typically assume references that are either clean or geometrically well-aligned or they rely on auxiliary semantic priors to regulate the denoising process. Under realistic satellite conditions where reference images are degraded and affected by geometric distortion, maintaining reliable structural correspondence remains challenging, which can lead to compromised reconstruction fidelity.

In contrast, the proposed SFSR framework is explicitly designed for realistic NFOV–WFOV satellite settings. Rather than emphasizing precise alignment or unconstrained generative synthesis, SFSR focuses on frequency-aware reference utilization, enabling robust fusion of degraded NFOV references with WFOV inputs under severe degradation and large scaling factors.

III. Methodology

A. Framework Overview

The proposed SFSR framework is a conditional diffusion model that enhances WFOV satellite imagery using degraded NFOV reference images. Unlike conventional RefSR methods that assume clean or well-aligned references [6], [7], [8], [9], [10], [36], [38], SFSR is explicitly designed for realistic cross-sensor settings, where NFOV references commonly exhibit blur, noise, and mild geometric misalignment.

As shown in Fig. 2, SFSR employs a shared encoder to extract feature representations from both the WFOV LR input (X_{LR}) and the degraded NFOV reference (X_{Ref}), embedding them into a common feature space. These features are then processed by the S2CAM, which provides structured reference guidance during the reverse diffusion process.

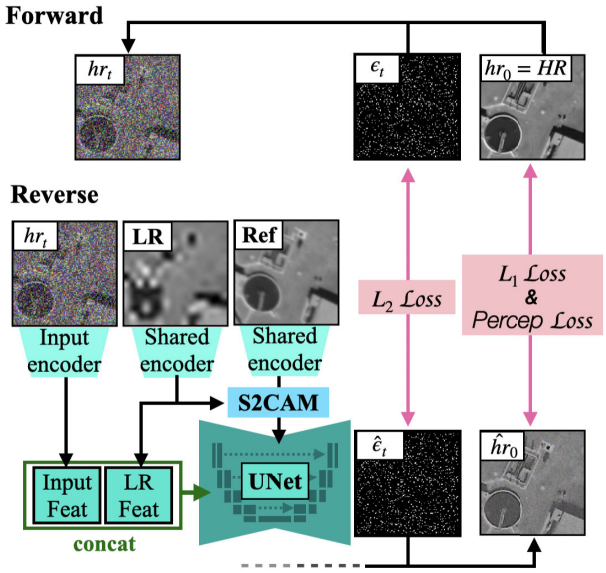


Fig. 2. Overview of SFSR. The model super-resolves WFOV (LR) images using degraded NFOV references (Ref) through an S2CAM-guided reverse diffusion process.

During reconstruction, S2CAM progressively refines the latent representation hr_t toward an HR output by incorporating reference information in a frequency-aware manner, rather than relying on explicit spatial feature alignment. This design allows the model to exploit complementary reference details while alleviating the impact of geometric misalignment.

Frequency-aware reference utilization in SFSR is realized through two dedicated submodules. The SSP decomposes degraded NFOV reference features into frequency-specific components, separating global structural information from fine-scale details so that reference cues can be selectively incorporated during cross-attention. The NSP further analyzes the high-frequency components by identifying responses dominated by sensor noise or artifacts and separating them from edge-relevant structural signals. Together, SSP and NSP convert degraded reference features into more reliable and interpretable representations, enabling stable and controlled reference guidance under realistic NFOV degradation.

B. Spectral Splitter

NFOV reference images often contain spectrally distorted or spatially shifted signals caused by sensor optics, temporal gaps, and atmospheric effects [39], [40]. Such degradations make it difficult to directly utilize reference features for super-resolution. The SSP addresses this challenge by decomposing reference features into frequency-specific components, enabling more controlled and reliable reference utilization under degraded conditions.

Unlike earlier approaches that rely on fixed frequency partitioning or global attention weighting [32], [33], [34], SSP performs a learnable, channel-wise frequency decomposition that adapts to the degradation characteristics of each reference image. This design allows reference features to be reorganized

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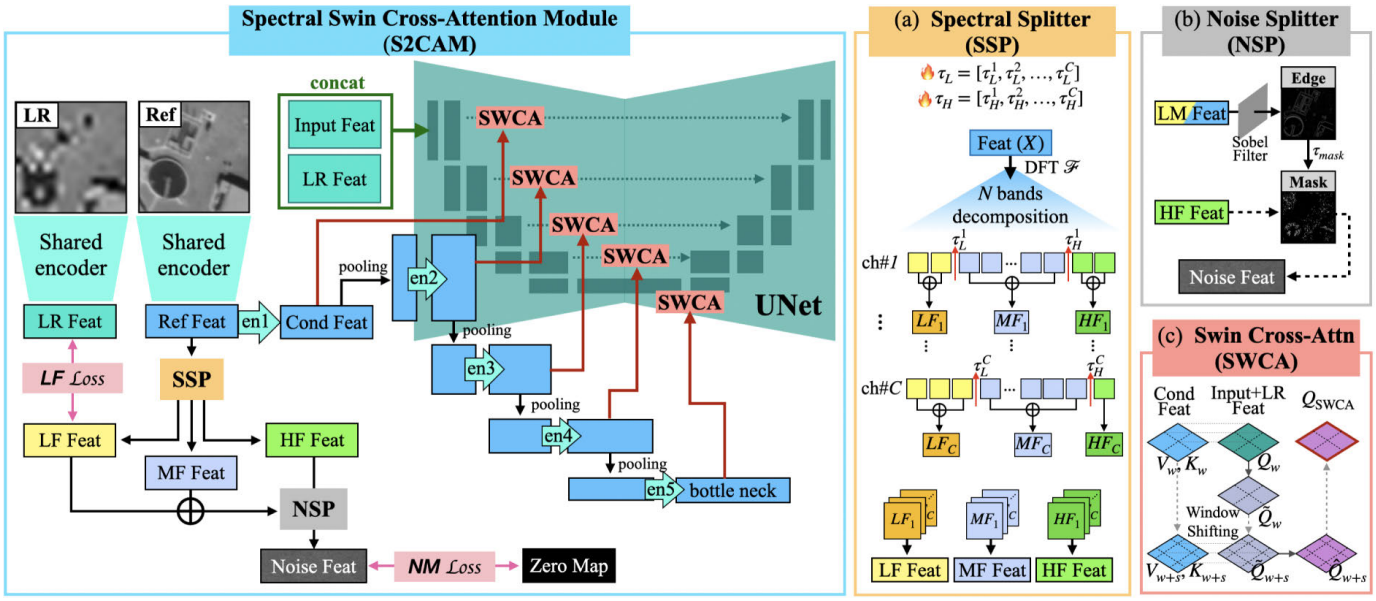


Fig. 3. S2CAM in SFSR. (a) SSP and (b) NSP refine features to overcome degradation in NFOV references, while SWCA integrates the refined reference information during the reverse diffusion process.

into representations that are better suited for frequency-aware reference feature utilization.

As illustrated in Fig. 3(a), a 2-D discrete Fourier transform (DFT) is applied independently to each reference feature channel c

$$\hat{X}_{\text{Ref}}^{(c)} = \mathcal{F}\left(X_{\text{Ref}}^{(c)}\right) = \sum_{n=1}^{N} \hat{X}_{\text{Ref}}^{(c,n)} \tag{1}$$

where $\hat{X}_{\text{Ref}}^{(c)}$ denotes the frequency-domain coefficients. Two learnable thresholds, τ_L^c and τ_H^c, partition the spectrum into low-, mid-, and high-frequency bands

$$\hat{X}_{\text{Ref}}^{\text{LF}} := \sum_{n<\tau_L^c} \hat{X}_{\text{Ref}}^{(c,n)} \tag{2}$$

$$\hat{X}_{\text{Ref}}^{\text{MF}} := \sum_{\tau_L^c \le n \le \tau_H^c} \hat{X}_{\text{Ref}}^{(c,n)} \tag{3}$$

$$\hat{X}_{\text{Ref}}^{\text{HF}} := \sum_{n>\tau_H^c} \hat{X}_{\text{Ref}}^{(c,n)}. \tag{4}$$

Each frequency band is transformed back to the spatial domain via inverse Fourier transform

$$X_{\text{Ref}}^{b} = \mathcal{F}^{-1}\left(\hat{X}_{\text{Ref}}^{b}\right), \quad b \in \{\text{LF}, \text{MF}, \text{HF}\}. \tag{5}$$

These frequency-specific representations correspond to different structural scales. Low-frequency components capture global layout and coarse structure, mid-frequency (MF) components encode object boundaries and fine spatial details, and high-frequency components mainly contain sharp responses and noise. By explicitly separating reference features in this manner, SSP constructs representations that facilitate selective and stable reference guidance in the presence of NFOV degradation.

C. Noise Splitter

High-frequency reference features $X_{\text{Ref}}^{\text{HF}}$ often contain responses that are unreliable for reconstruction, originating from sensor noise, optical artifacts, or fine-scale fluctuations that lack structural support. Rather than performing explicit denoising, the NSP aims to identify and separate such noise-dominated responses from structurally meaningful high-frequency information.

As illustrated in Fig. 3(b), NSP evaluates the reliability of high-frequency responses using more stable low-frequency and MF reference features. The low-frequency and MF components are first aggregated as

$$X_{\text{Ref}}^{\text{LM}} = X_{\text{Ref}}^{\text{LF}} + X_{\text{Ref}}^{\text{MF}} \tag{6}$$

which provides a robust representation of the underlying scene structure. A Sobel operator $\mathcal{S}(\cdot)$ is then applied to extract edge responses

$$E = \mathcal{S}\left(X_{\text{Ref}}^{\text{LM}}\right), \quad M = \text{Ind}\left(E < \tau_{\text{mask}}\right) \tag{7}$$

where τ_{mask} is an edge mask threshold and M denotes a binary mask indicating regions without sufficient edge evidence.

This mask is used to isolate noise-dominated high-frequency responses

$$X_{\text{Ref}}^{\text{Noise}} = X_{\text{Ref}}^{\text{HF}} \odot M \tag{8}$$

where $\odot$ denotes the element-wise multiplication. High-frequency responses identified by M are treated as unreliable and excluded from reference feature fusion.

By separating structurally unsupported high-frequency responses instead of suppressing all high-frequency content, NSP ensures that only edge-consistent and structurally meaningful reference details are propagated to subsequent cross-attention operations. This design enables stable and interpretable reference utilization under degraded NFOV conditions.

D. Spectral Swin Cross-Attention Module

The S2CAM integrates the refined NFOV reference features produced by SSP and NSP into the diffusion backbone. Rather

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than enforcing explicit feature alignment, S2CAM enables structured reference utilization through a two-stage shifted-window cross-attention (SWCA) mechanism [22].

As illustrated in Fig. 3(c), SWCA computes cross-attention within local windows in the first stage, allowing the model to incorporate reference cues that are spatially coherent with the WFOV features. In the second stage, the window positions are shifted, enabling information exchange across neighboring regions and facilitating broader contextual interaction without resorting to global attention

$$\hat{Q}^{(l)}_{w+s} = \mathrm{Attn}\left(\tilde{Q}^{(l)}_{w+s}, K^{(l)}_{w+s}, V^{(l)}_{w+s}\right) \tag{9}$$

$$Q^{(l)}_{\mathrm{SWCA}} = \mathrm{RS}\left(\mathrm{Merge}\left(\hat{Q}^{(l)}_{w+s}\right)\right) \tag{10}$$

where $\tilde{Q}$ denotes the WFOV query features and K and V are derived from the spectrally refined NFOV reference features.

By restricting attention to local regions before progressively propagating information across shifted windows, S2CAM incorporates reference features in a spatially controlled manner. This design reduces sensitivity to local misalignment and partial overlap that commonly arise in multisensor satellite imagery. Consequently, S2CAM effectively transfers frequency-refined and structurally supported reference cues into the reverse diffusion process while preserving spatial consistency.

E. Objective Function

The total training objective combines five complementary loss terms to jointly address diffusion noise prediction, image reconstruction, perceptual quality, and frequency-aware reference utilization

$$\mathcal{L}_{\mathrm{total}} = c_1\mathcal{L}_2 + c_2\mathcal{L}_1 + c_3\mathcal{L}_{\mathrm{VGG}} + c_4\mathcal{L}_{\mathrm{LF}} + c_5\mathcal{L}_{\mathrm{NM}}. \tag{11}$$

1) L_2 Loss for Noise Prediction: This term optimizes the reverse diffusion process by predicting the Gaussian noise ϵ added to the latent variable hr_t

$$\mathcal{L}_2 = \left\|\mathrm{UNet}_\theta\left(I_{\mathrm{LR}}, I_{\mathrm{Ref}}, hr_t, \bar{\alpha}_t\right) - \epsilon\right\|_2 \tag{12}$$

$$hr_t = \sqrt{\bar{\alpha}_t}\, hr_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \tag{13}$$

2) L_1 Reconstruction Loss: Enforce pixel-wise consistency between the reconstructed output and the ground-truth HR image

$$\mathcal{L}_1 = \left\|\hat{hr}_0 - hr_0\right\|_1. \tag{14}$$

3) Perceptual Loss: A VGG-based perceptual loss [41] is used to encourage visually plausible reconstruction

$$\mathcal{L}_{\mathrm{VGG}} = \left\|\phi\left(\hat{hr}_0\right) - \phi\left(hr_0\right)\right\|_1 \tag{15}$$

where $\phi(\cdot)$ denotes feature activations extracted from a pre-trained VGG network.

4) Low-Frequency Alignment Loss: It constrains the low-frequency component of the reference features to remain consistent with the global feature of the WFOV input

$$\mathcal{L}_{\mathrm{LF}} = \left\|X_{\mathrm{LR}} - X^{\mathrm{LF}}_{\mathrm{Ref}}\right\|_2. \tag{16}$$

This loss ensures that the low-frequency content of the reference remains consistent with the overall structure of the WFOV input so that reference-based fusion focuses on adding finer details rather than altering global structure.

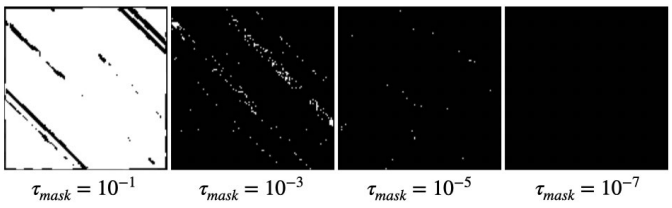


Fig. 4. Visualization of edge-based masking results under different edge mask thresholds (τ_{mask}). White regions indicate high-frequency areas excluded due to insufficient edge support, while black regions denote edge-consistent regions preserved for reference utilization.

5) Noise Minimization Loss: It penalizes noise-dominated high-frequency reference responses identified by the NSP

$$\mathcal{L}_{\mathrm{NM}} = \left\|X^{\mathrm{Noise}}_{\mathrm{Ref}}\right\|_2. \tag{17}$$

This term discourages unreliable high-frequency signals from influencing reference fusion, improving the stability of spectral redistribution across degraded NFOV inputs.

F. Mechanism of Feature Redistribution

SFSR regulates reference utilization across frequency bands through learning rather than predefined spectral assignments. Instead of fixing the role of each frequency band, the model learns to redistribute reference features according to their structural reliability during training. This learning-driven behavior arises from the joint interaction of the shared encoder, learnable spectral boundaries, and edge-guided masking, which collectively govern how reference information contributes to reconstruction in the reverse diffusion process.

1) Edge Mask Threshold Selection: The edge mask threshold τ_{mask} controls how high-frequency reference features are selectively filtered in the NSP. NSP computes a binary mask from the Sobel magnitude of the combined low-frequency and MF features and uses this mask to gate high-frequency responses. As a result, τ_{mask} determines which high-frequency regions lack sufficient edge support and are treated as noise-dominated.

Fig. 4 visualizes the resulting edge-based masks under different threshold values. When τ_{mask} is large (e.g., 10^{-1}), the masked regions (white) become overly aggressive, excluding not only noise-dominated responses but also structurally meaningful edges. In contrast, very small thresholds (e.g., 10^{-7}) produce weak masking, allowing unreliable high-frequency artifacts to remain in the reference features.

An intermediate threshold, $\tau_{\mathrm{mask}} = 10^{-5}$, offers a balanced behavior; it removes high-frequency responses that are not supported by edges while retaining most meaningful structural details inferred from low-frequency and MF features. As a result, reference features are used more stably and feature redistribution remains consistent during training. We therefore adopt $\tau_{\mathrm{mask}} = 10^{-5}$ in all experiments. We note that τ_{mask} is a data-dependent parameter and may require adjustment under different noise or sensor conditions, which we leave for future work.

2) Frequency-Aware Reference Utilization: To analyze how SFSR exploits NFOV reference information across frequency

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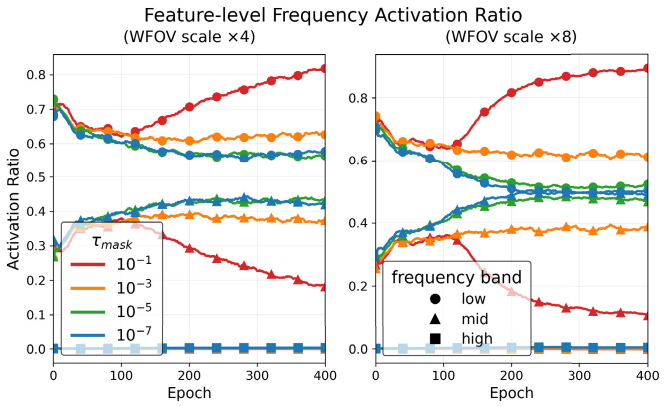


Fig. 5. Frequency activation ratio of NFOV reference features during training, showing how reference utilization is redistributed across frequency bands.

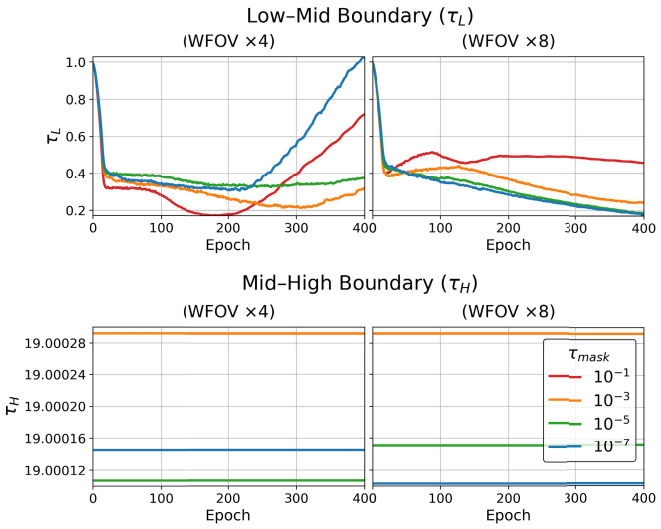


Fig. 6. Evolution of the learned spectral boundaries (τ_L and τ_H) during training for ×4 and ×8 WFOV settings. The curves are obtained by averaging results over five independent training runs.

bands, we examine the feature-level *frequency activation ratio*. This metric measures the relative contribution of reference features in the low-, mid-, and high-frequency bands by computing the average activation magnitude within each band after spectral decomposition. Specifically, for a given reference feature map, we compute the mean absolute activation of each frequency component and normalize them to obtain their relative ratios.

Fig. 5 shows consistent trends for both ×4 and ×8 settings. The edge mask threshold τ_{mask} plays a central role in shaping this frequency-wise utilization. When τ_{mask} is large, a larger portion of high-frequency reference responses is identified as noise-dominated by the NSP and is subsequently penalized through the NM Loss. As a result, reference utilization becomes dominated by low-frequency components. As τ_{mask} decreases, fewer high-frequency responses are separated as noise, allowing mid- and high-frequency components to contribute more actively during reference fusion. This behavior is consistently observed across both scaling factors.

This trend directly reflects the design intent of SFSR. Since the WFOV (LR) input already provides reliable low-frequency structural information, the shared encoder and SSP with LF Loss encourage reduced reliance on redundant low-frequency reference features and promote the use of mid- and high-frequency components that complement the information missing in the WFOV input. This redistribution becomes clearly apparent when $\tau_{\text{mask}} \leq 10^{-3}$, where reference utilization shifts away from low-frequency components toward increased mid- and high-frequency contributions.

The learned spectral boundaries (τ_L and τ_H) shown in Fig. 6 emerge jointly with the shared encoder to support frequency-aware reference utilization. These boundaries are not manually specified but are optimized as auxiliary variables during training. Because τ_L and τ_H are learned independently for each feature channel, the values reported in Fig. 6 represent the channel-averaged boundaries. Despite scale-dependent differences in their trajectories, several consistent patterns can be observed. In particular, the mid–high boundary τ_H converges near the highest frequency band for both ×4 and ×8, indicating that a single high-frequency band among the 20 available bands is sufficient to aggregate noise-dominated responses for subsequent suppression.

The behavior of τ_L is more sensitive to the WFOV scale. At ×8, where the WFOV input is strongly dominated by low-frequency content, increasing τ_{mask} leads to a higher activation ratio of low-frequency reference components, as shown in Fig. 5. Under this condition, the model relies more heavily on low-frequency reference information, and τ_L converges to higher values accordingly. At ×4, where the WFOV input retains more mid- and high-frequency information, low-frequency reference components may also carry partial detail cues, leading to more complex and less monotonic τ_L dynamics during training. However, despite these scale-dependent variations, the frequency activation ratios remain consistent across both settings. In all cases, SFSR reduces reliance on redundant low-frequency reference information, emphasizes informative MF structures, and confines high-frequency noise to a dedicated suppression band.

3) Spectral Manifestation in Feature Space: The effect of the proposed frequency-aware reference utilization is illustrated in Fig. 7, which shows feature heat maps (top row) and corresponding Fourier magnitude spectra (bottom row). Before training [Fig. 7(a)], the spectral structure is poorly organized across frequency bands. Low-frequency energy is broadly diffused around the center, while MF components lack clear band-wise separation and instead exhibit dispersed cross-shaped patterns. High-frequency responses are sparsely concentrated near the spectral corners, where signals are more likely to be dominated by noise or unstable artifacts.

This behavior is reflected in the feature heat maps, where NFOV reference features closely resemble WFOV input features and mainly capture coarse, low-frequency structures. Consequently, although reference matching appears straightforward, the reference features contribute little complementary information beyond what is already present in the WFOV input.

After training [Fig. 7(b)], the spectral structure becomes clearly organized across frequency bands. Low-frequency (LF) features show a compact and sharply concentrated energy

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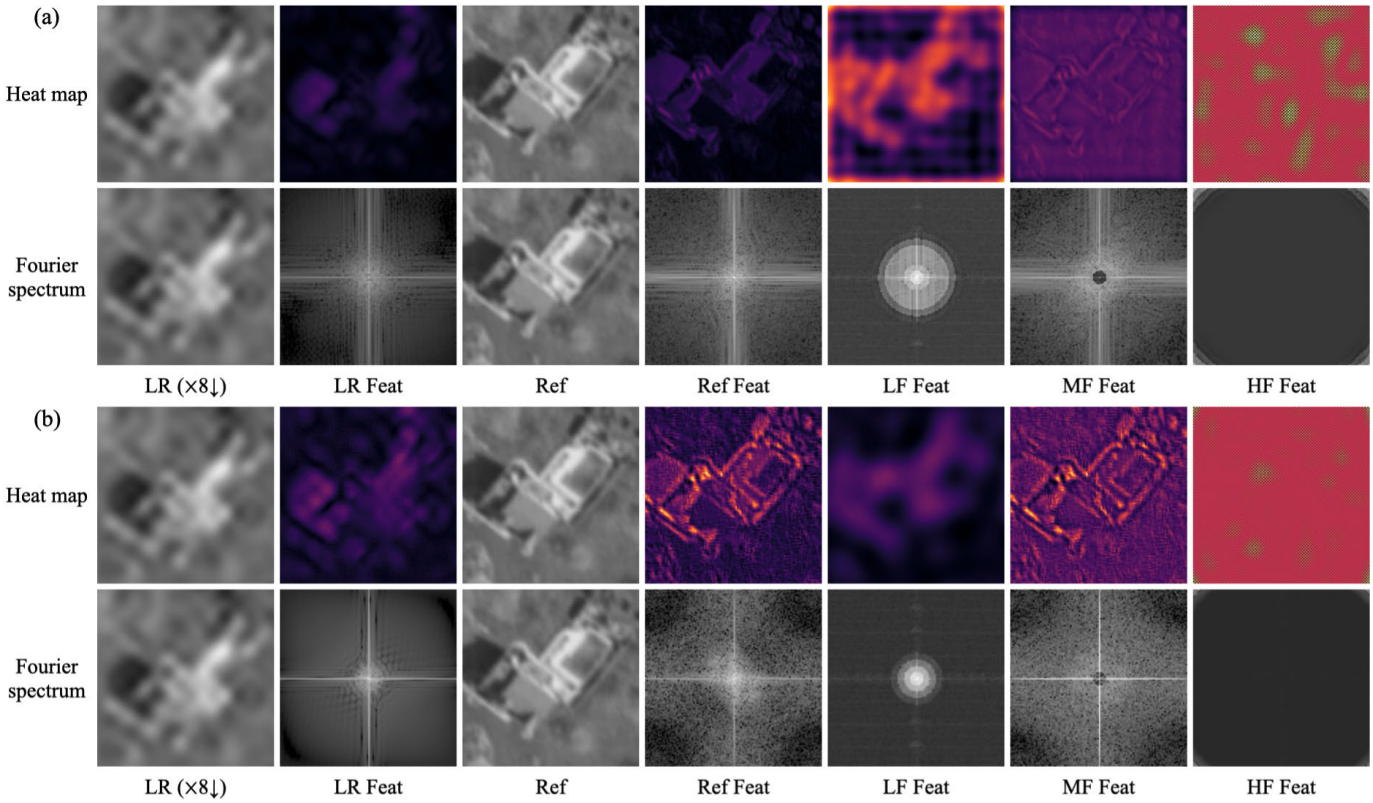


Fig. 7. Frequency-domain visualization of WFOV (LR) and reference NFOV reference (Ref) feature decomposition using frequency-domain heat maps and Fourier magnitude spectra. (a) Before training, where SSP, NSP, LF loss, and NM loss are applied, but the shared encoder and spectral boundaries (τ_L and τ_H) remain at their initial states. (b) After training, showing redistributed spectra induced by the learned shared encoder and spectral boundaries.

distribution at the center. The close match between the LR and LF spectra confirms that global information is mainly handled in the low-frequency band. MF features form a ring-shaped energy distribution with sharp cross-shaped responses, indicating that complementary structural and textural details are selectively encoded in this range. In contrast, high-frequency (HF) components are strongly suppressed, with energy at the four corners largely eliminated, suggesting that these bands primarily absorb noise-dominated responses.

The feature heat maps show that the NFOV reference features now include fine details that are not present in the WFOV input, indicating that the reference provides genuinely new information rather than duplicating existing content. Overall, SFSR learns to use reference information more effectively by reducing unnecessary low-frequency overlap, focusing on useful MF details, and isolating high-frequency noise into a separate suppression band.

IV. Experiments

A. Synthetic Degradation Pipeline

Unlike conventional super-resolution studies that primarily model degradations using blur and additive noise alone [41], [42], our synthetic degradation pipeline is designed to explicitly reflect the optical degradation and geometric misalignment commonly observed in degraded NFOV reference images. This consideration is critical in remote sensing scenarios, where reference images are often acquired under different focal lengths, viewing geometries, or sensor configurations, leading to systematic yet subtle misalignment between WFOV inputs and NFOV references.

To approximate such conditions, the proposed pipeline jointly models optical degradation and geometric misalignment. Both the WFOV (LR) image and the NFOV reference (Ref) are generated from the same HR scene to preserve underlying structural consistency, while allowing pixel-level discrepancies to arise through controlled degradation. All images are resized or cropped to 192×192, and larger scenes are divided into overlapping patches with a 64-pixel stride. The central 128×128 region of each patch is used as the HR target. WFOV inputs are obtained by downsampling the HR images by factors of $\times 4$ or $\times 8$, followed by bicubic upsampling to 128×128 to ensure spatial compatibility with the degraded NFOV references.

Degradation of NFOV reference images is simulated through a combination of physically motivated optical blur and geometric perturbation. Specifically, diffraction blur caused by limited aperture is modeled using a Bessel kernel (21×21, aperture 0.5), while direction-dependent degradation characteristic of narrow-aperture imaging is introduced using randomly oriented sinc kernels ($\theta \sim U(0, 2\pi)$). To further reflect realistic sensing conditions, additive uniform noise $\epsilon \sim U(-0.03, 0.03)$ is applied, along with random homography perturbations of up to ± 5 pixels at the image corners. A final crop to 128×128 is then performed.

Although this synthetic NFOV–WFOV setting cannot fully capture the complexity of real multisensor satellite measurements, it is designed to cover a broad and controlled range

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of lens-induced blur, noise, and geometric misalignment. By explicitly modeling these degradations, the proposed synthetic degradation pipeline provides a structured training distribution that enables robust and stable reference utilization under degraded NFOV conditions.

B. Training Details and Datasets

SFSR is trained with a batch size of 16 under a cosine noise schedule over 2000 diffusion steps following the DDPM framework [21]. The learning rate is set to 10^{-3} for the learnable spectral boundaries in the SSP and 10^{-4} for all remaining parameters and is decayed to 5×10^{-4} and 5×10^{-5}, respectively. The NSP employs a fixed edge mask threshold $\tau_{\text{mask}} = 10^{-5}$ to identify noise-dominated high-frequency responses, which are subsequently penalized through the NM Loss. The overall training objective combines five complementary loss terms with weights $c_1 = 1.0$, $c_2 = 0.5$, $c_3 = 0.02$, $c_4 = 0.01$, and $c_5 = 10^6$, balancing diffusion noise prediction, structural fidelity, perceptual quality, and frequency-aware reference utilization.

Training and evaluation are conducted at $\times 4$ and $\times 8$ upscaling factors on an NVIDIA RTX 3090 GPU using PyTorch. During inference, we adopt DDIM sampling [24] with 200 reverse diffusion steps to achieve efficient and stable reconstruction.

For synthetic training and controlled evaluation, we use the UC Merced Land Use Dataset (UCM-LUD) [18], which contains 2100 aerial images (2000 for training and 100 for evaluation) with diverse land-cover patterns. This dataset is well suited for learning super-resolution under synthetic WFOV–NFOV degradation. To further assess cross-domain generalization, we evaluate downstream performance on DOTA [19] (455 images across 15 object classes) for object detection and UV6K [35] (500 vehicle mask images) for small-object segmentation. These experiments verify that SFSR improves both detection and segmentation performance under degraded NFOV reference conditions.

In addition to synthetic data, we evaluate SFSR on a real cross-sensor dataset, NAIP [53], [54], to examine generalization under practical imaging conditions. In this setting, images with a ground sampling distance (GSD) of 1.0 m are treated as NFOV reference images, while images with a GSD of 0.6 m are downsampled to generate WFOV (LR) inputs. Specifically, the 0.6-m images are downsampled by factors of $\times 4$ and $\times 8$, corresponding to effective GSDs of 2.4 and 4.8 m, respectively. A total of 1000 paired NFOV–WFOV samples are used for fine-tuning, and an additional 1000 pairs are reserved for evaluation. Although the WFOV images are synthetically generated through downsampling, this dataset provides real cross-sensor reference characteristics and complements the synthetic experiments.

V. Results

A. Evaluation on Synthetic Degradation Data

We evaluate SFSR against state-of-the-art SISR and RefSR methods on three benchmark datasets, UCM-LUD [18], DOTA [19], and UV6K [35], under the proposed synthetic WFOV–NFOV degradation setting. All experiments are conducted at $\times 4$ and $\times 8$ upscaling factors. For a fair comparison, all baseline models are trained or fine-tuned using the same synthetic degradation pipeline, and their input and output configurations are consistently adjusted to support both scaling factors.

Performance is evaluated using peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM) [17], learned perceptual image patch similarity (LPIPS) [15], and Fréchet inception distance (FID) [51], which together assess reconstruction accuracy, structural consistency, perceptual quality, and distribution-level realism. All reported results are averaged over five independent training runs with different random seeds to reduce variance caused by initialization and training stochasticity.

The compared methods include representative SISR approaches, namely, LANCZOS interpolation [26], SwinIR [12], SRGAN [11], SR3 [1], EDiffSR [47], and BSRGAN [45], as well as advanced RefSR models including TTSR [8], DATSR [10], KeDuSR [36], and RefDiff [38]. These baselines cover a broad range of interpolation-based, convolutional, transformer-based, and diffusion-based designs, as well as diverse reference utilization strategies.

To ensure a controlled comparison, the SR3 baseline is implemented using the same UNet [13] backbone and diffusion settings as SFSR, with all reference-related components removed. For RefDiff, which originally relies on additional reference region masks or semantic annotations, we adopt a unified setting in which the entire reference image is used as guidance during both training and evaluation. This choice avoids extra supervision and ensures fair comparison across methods while also constituting a conservative evaluation setting for RefDiff, as no explicit reference region cues are provided.

1) Quantitative Comparison: Under the synthetic NFOV–WFOV degradation setting, as shown in Tables I and II, SFSR consistently outperforms all competing methods across all datasets and scaling factors. It achieves the highest PSNR and SSIM while also recording the lowest LPIPS and FID scores, indicating superior performance in both pixel-level accuracy and perceptual quality. This advantage becomes more evident at $\times 8$, where stronger resolution loss and heavier reference degradation make reconstruction more challenging.

SISR methods, such as SwinIR, SRGAN, SR3, EDiffSR, and BSRGAN, rely only on the WFOV input. As a result, they struggle to recover fine structures under severe degradation. While diffusion-based SISR methods like SR3 and EDiffSR sometimes produce visually pleasing results, they often show lower pixel-wise accuracy and require substantially higher inference time, which limits their practicality for remote sensing applications.

RefSR methods, including TTSR, DATSR, KeDuSR, RefDiff, and SFSR, mitigate this limitation by exploiting NFOV references. Among them, DATSR is the strongest nondiffusion baseline and shows relatively stable performance under moderate misalignment. However, many RefSR methods become unstable when the reference image is strongly

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TABLE I

QUANTITATIVE COMPARISON ON BENCHMARK DATASETS FOR ×4 SUPER-RESOLUTION UNDER SYNTHETIC NFOV–WFOV DEGRADATION, COMPARING SFSR WITH STATE-OF-THE-ART SISR AND REFSR METHODS

Dataset	UCM-LUD			DOTA			UV6K			params
Metric	PSNR↑ / SSIM↑	LPIPS↓ / FID↓	time(s)↓	PSNR↑ / SSIM↑	LPIPS↓ / FID↓	time(s)↓	PSNR↑ / SSIM↑	LPIPS↓ / FID↓	time(s)↓	(M)
Ref	15.31 / 0.174	0.622 / 248.46	-	21.06 / 0.619	0.173 / 79.60	-	16.47 / 0.391	0.138 / 56.55	-	-
LR(×4 ↓)	22.76 / 0.605	0.472 / 391.60	-	27.16 / 0.619	0.107 / 70.65	-	21.61 / 0.648	0.016 / 36.80	-	-
LANCZOS [26]	24.07 / 0.652	0.550 / 239.58	-	27.69 / 0.801	0.091 / 63.38	-	22.27 / 0.652	0.011 / 32.06	-	-
SwinIR [12]	25.16 / 0.720	0.398 / 244.92	0.11	28.44 / 0.825	0.033 / 46.62	1.00	23.21 / 0.715	0.005 / 43.88	1.59	11.82
SRGAN [11]	24.41 / 0.683	0.375 / 266.41	0.09	28.55 / 0.829	0.050 / 49.25	0.82	22.95 / 0.688	0.006 / 29.61	1.62	22.15
SR3 [1]	22.68 / 0.594	0.247 / 169.34	1.58	27.57 / 0.798	0.026 / 87.13	12.17	21.97 / 0.636	0.005 / 16.15	19.71	37.94
EDiffSR [47]	25.89 / 0.676	0.324 / 169.76	2.21	28.42 / 0.816	0.031 / 41.99	16.34	22.86 / 0.679	0.004 / 38.99	29.22	26.79
BSRGAN [45]	24.90 / 0.698	0.331 / 213.20	0.03	27.67 / 0.767	0.067 / 43.15	0.19	21.04 / 0.638	0.017 / 35.93	0.35	16.61
TTSR [8]	24.50 / 0.682	0.352 / 229.33	0.14	28.80 / 0.829	0.039 / 46.16	0.67	23.01 / 0.689	0.004 / 35.56	1.13	6.25
DATSR [10]	25.13 / 0.742	0.346 / 220.32	0.17	29.19 / 0.856	0.036 / 34.26	0.97	23.48 /0.744	0.004 / **19.19**	1.49	10.51
KeDuSR [36]	25.16 / 0.713	0.376 / 249.52	0.02	27.83 / 0.779	0.088 / 73.57	0.21	23.34 / 0.720	0.005 / 31.84	0.37	5.63
RefDiff [38]	21.28 / 0.531	0.361 / 219.41	11.08	23.29 / 0.665	0.059 / 36.05	123.11	21.07 / 0.594	0.020 / 25.99	171.97	116.74
SFSR (20 steps)	27.54 / 0.799	0.209 / 154.14	0.39	30.01 / 0.863	0.016 / 27.11	2.31	24.66 / 0.780	0.002 / 22.04	3.40	69.96
SFSR (200 steps)	**27.81 / 0.813**	**0.189 / 150.55**	3.08	**30.23 / 0.871**	**0.015 / 24.85**	20.01	**24.97 / 0.784**	**0.002** / 21.77	28.28	69.96

TABLE II

QUANTITATIVE COMPARISON ON BENCHMARK DATASETS FOR ×8 SUPER-RESOLUTION UNDER SYNTHETIC NFOV–WFOV DEGRADATION, COMPARING SFSR WITH STATE-OF-THE-ART SISR AND REFSR METHODS

Dataset	UCM-LUD			DOTA			UV6K			params
Metric	PSNR↑ / SSIM↑	LPIPS↓ / FID↓	time(s)↓	PSNR↑ / SSIM↑	LPIPS↓ / FID↓	time(s)↓	PSNR↑ / SSIM↑	LPIPS↓ / FID↓	time(s)↓	(M)
Ref	15.31 / 0.174	0.622 / 248.46	-	21.06 / 0.619	0.173 / 79.60	-	16.47 / 0.391	0.138 / 56.55	-	-
LR(×8 ↓)	20.30 / 0.376	0.616 / 406.26	-	24.17 / 0.707	0.346 / 158.49	-	19.75 / 0.488	0.137 / 110.01	-	-
LANCZOS [26]	21.09 / 0.424	0.755 / 396.04	-	26.00 / 0.722	0.298 / 155.04	-	20.02 / 0.481	0.237 / 84.46	-	-
SwinIR [12]	21.43 / 0.458	0.693 / 388.65	0.11	26.01 / 0.726	0.227 / 125.94	1.01	20.37 / 0.506	0.138 / 64.35	1.57	11.82
SRGAN [11]	21.48 / 0.454	0.612 / 365.95	0.08	26.26 / 0.729	0.197 / 128.31	0.82	20.42 / 0.508	0.115 / 61.37	1.66	22.15
SR3 [1]	20.07 / 0.344	0.410 / 241.03	1.59	25.36 / 0.694	0.130 / 135.00	12.01	19.83 / 0.461	0.100 / 65.35	19.33	37.94
EDiffSR [47]	23.12 / 0.471	0.644 / 262.77	2.25	25.99 / 0.718	0.264 / 108.61	15.99	20.05 / 0.479	0.212 / 75.93	30.51	26.79
BSRGAN [45]	21.29 / 0.457	0.523 / 221.63	0.03	25.10 / 0.701	0.105 / 90.38	0.19	19.29 / 0.466	0.121 / 59.55	0.39	16.61
TTSR [8]	21.36 / 0.458	0.500 / 340.51	0.14	26.18 / 0.731	0.171 / 120.56	0.66	20.46 / 0.514	0.104 / 64.63	1.13	6.25
DATSR [10]	22.30 / 0.556	0.519 / 319.13	0.17	27.49 / 0.824	0.049 / 43.72	0.99	20.96 /0.582	0.067 / 40.19	1.48	10.51
KeDuSR [36]	21.71 / 0.481	0.552 / 330.91	0.02	25.59 / 0.705	0.241 / 130.49	0.21	20.54 / 0.528	0.111 / 60.16	0.36	5.63
RefDiff [38]	19.58 / 0.486	0.391 / 224.81	10.99	22.77 / 0.646	0.075 / 54.75	120.19	19.10 / 0.411	0.035 / 40.28	169.45	116.74
SFSR (20 steps)	25.39 / 0.762	0.237 / 162.72	0.39	27.59 / 0.823	0.037 / 45.12	2.36	22.15 / 0.707	0.022 / 32.07	3.55	69.96
SFSR (200 steps)	**25.83 / 0.762**	**0.199 / 162.90**	3.11	**28.21 / 0.828**	**0.036 / 43.07**	21.17	**22.94 / 0.704**	**0.017 / 29.06**	28.30	69.96

distorted. RefDiff, a diffusion-based RefSR method, can generate visually rich textures and often achieves favorable perceptual scores. However, this often comes at the cost of lower pixel-level accuracy and higher computational cost. In addition, RefDiff may hallucinate details that are not well supported by the LR input, which is undesirable in remote sensing tasks that require reliable structural accuracy. In contrast, SFSR achieves a better balance between perceptual quality and structural fidelity. By explicitly controlling reference utilization through frequency-aware decomposition and noise-aware handling, SFSR improves visual quality while preserving accurate scene structure. As a result, it consistently outperforms all RefSR baselines, including RefDiff, across all datasets and scaling factors.

Notably, SFSR remains effective even when the number of reverse diffusion steps is greatly reduced. The 20-step variant retains most of the performance gains and still surpasses all RefSR baselines, showing that high-quality reconstruction can be achieved without long sampling procedures.

2) Qualitative Comparison: Figs. 8 and 9 present qualitative comparisons among RefSR methods, focusing on reconstruction quality under degraded NFOV references. Each model reconstructs 128×128 patches, which are then stitched to form full-resolution images. In the NFOV setting, reference inputs suffer from optical blur, noise, and geometric distortion, leading to spatial misalignment and discontinuities across patches.

Conventional RefSR methods such as TTSR and KeDuSR rely strongly on accurate spatial alignment between the LR input and the reference image. When the reference is geometrically distorted, these methods tend to produce blurred edges, warped structures, and locally inconsistent textures. DATSR partially alleviates this issue by using deformable attention [10] to adaptively estimate correspondences between the LR and reference features. However, under severe distortion, it still exhibits noticeable contrast variation and texture inconsistency across different regions.

RefDiff, a diffusion-based RefSR method, generates visually rich and sharp textures, often achieving strong perceptual quality. However, its reconstructions frequently deviate from the ground truth in object shape and spatial arrangement. Such geometric inaccuracies reflect the generative nature of RefDiff, which emphasizes perceptual realism over strict structural fidelity. This behavior limits its applicability in remote sensing scenarios where accurate geometry and spatial consistency are essential.

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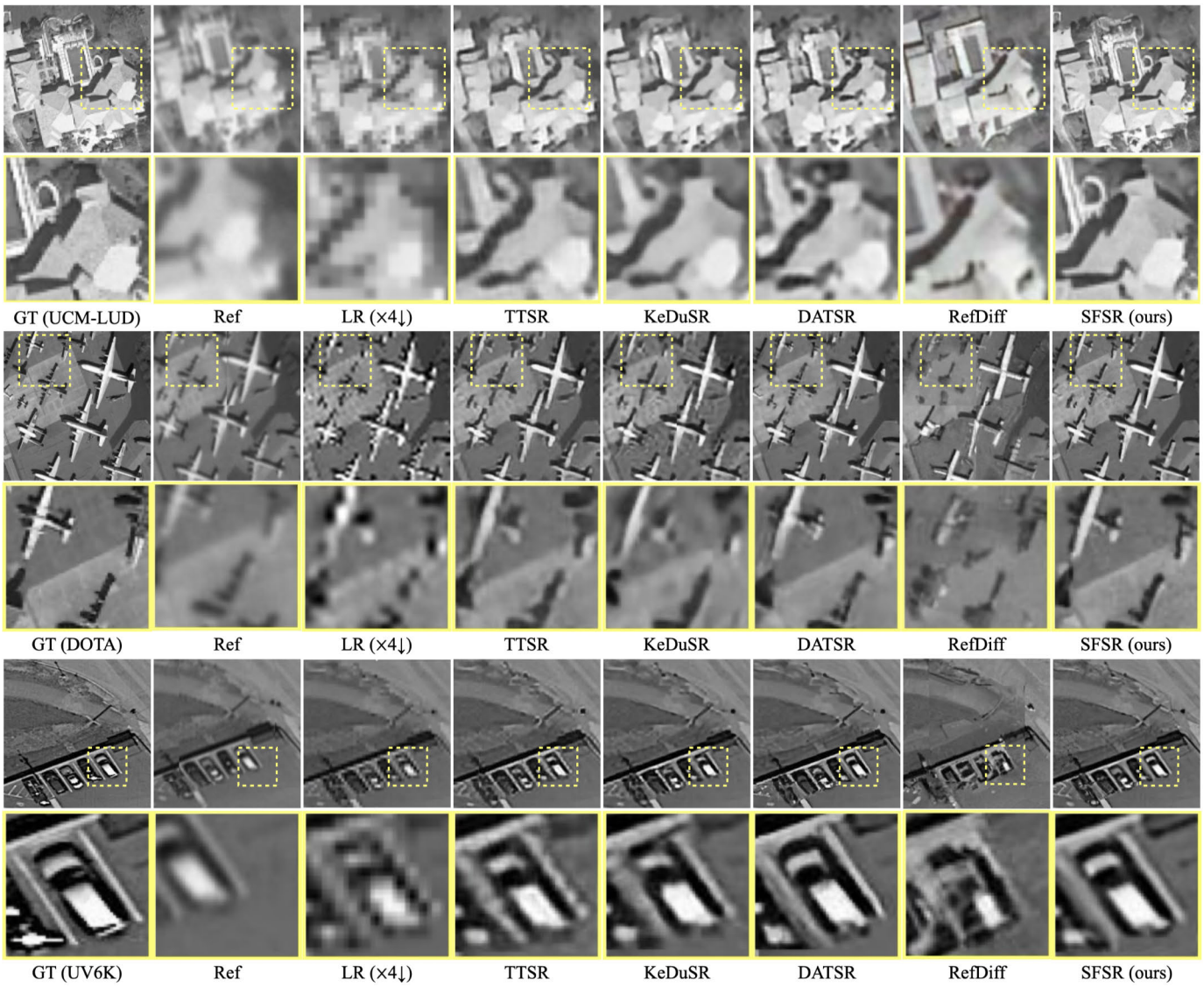


Fig. 8. Qualitative comparison of RefSR methods at ×4 WFOV scaling under synthetically degraded WFOV–NFOV conditions.

In contrast, SFSR produces reconstructions that are both sharp and structurally coherent. Fine details such as building boundaries, road edges, and vehicle shapes are consistently preserved, and stitched results remain globally stable without visible discontinuities. These qualitative results demonstrate that SFSR effectively exploits degraded reference information without introducing structural hallucinations, making it particularly suitable for remote sensing super-resolution tasks that demand high spatial fidelity.

B. Downstream Task Results

To evaluate the practical impact of super-resolution beyond image quality metrics, we assess the reconstructed images on downstream vision tasks, including object detection and semantic segmentation. In remote sensing applications, super-resolution is typically employed as a preprocessing step for higher level analysis, and improvements in visual appearance do not necessarily translate into better task performance.

Under the synthetic WFOV–NFOV degradation setting, we examine whether the structural fidelity and spatial consistency achieved by different methods lead to measurable improvements in downstream tasks. This evaluation allows us to distinguish methods that primarily enhance visual texture from those that reliably preserve task-relevant structural information under degraded reference conditions.

1) Object Detection on DOTA: We further evaluate the downstream performance of SFSR on the DOTA dataset [19] using a YOLOv5 [52] detector pretrained on 15 object classes. Detection accuracy is measured using $mAP_{50:95}$. As summarized in Table III, SFSR achieves the highest detection accuracy among all compared methods at both ×4 and ×8 WFOV super-resolution settings. At ×4, SFSR slightly outperforms DATSR, while at ×8, it shows a clearer margin, achieving 0.212 $mAP_{50:95}$ compared to 0.196 for DATSR. This performance gap becomes more pronounced as the WFOV resolution decreases and the reconstruction task becomes more challenging.

Table IV further reports class-wise detection results. At ×4 [Table IV(a)], SFSR achieves the highest $AP_{50:95}$ in 11 out of 15 object classes, and at ×8 [Table IV(b)], it leads in 13

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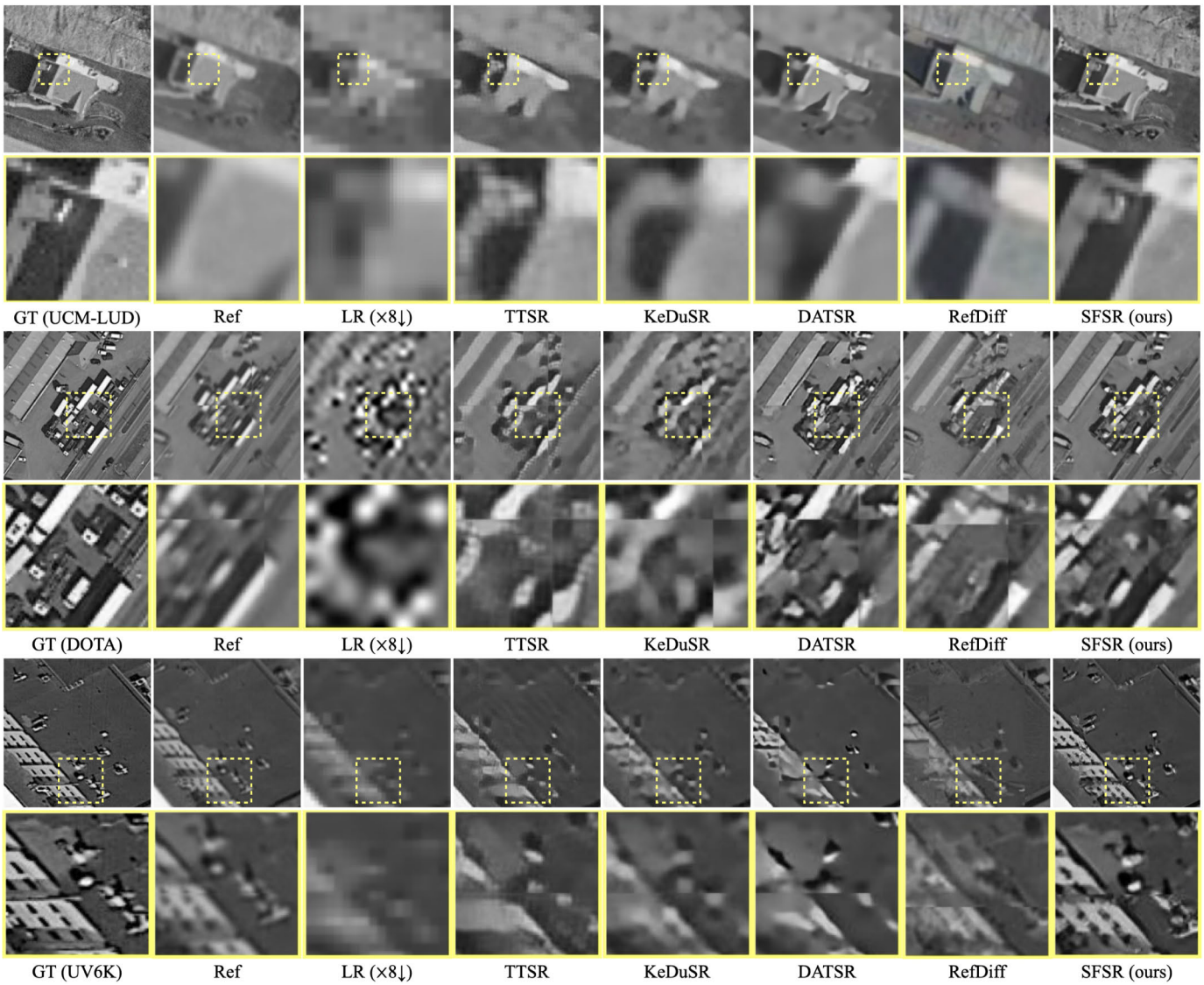


Fig. 9. Qualitative comparison of RefSR methods at ×8 WFOV scaling under synthetically degraded WFOV–NFOV conditions.

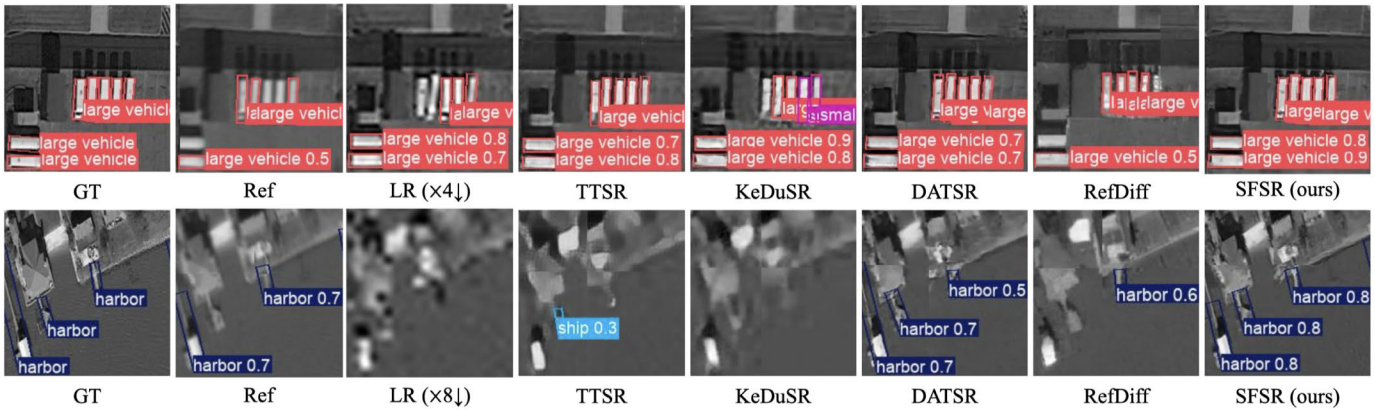


Fig. 10. Qualitative object detection results on the DOTA dataset using super-resolved images under synthetically degraded WFOV–NFOV conditions (first row: ×4 WFOV scaling and second row: ×8 WFOV scaling).

classes. In particular, for all seven military high-value target categories, namely, *bridge, harbor, helicopter, large vehicle, plane, ship,* and *small vehicle*, SFSR consistently ranks first or second across both scaling factors. These results indicate that SFSR maintains strong detection robustness across object categories and degradation levels.

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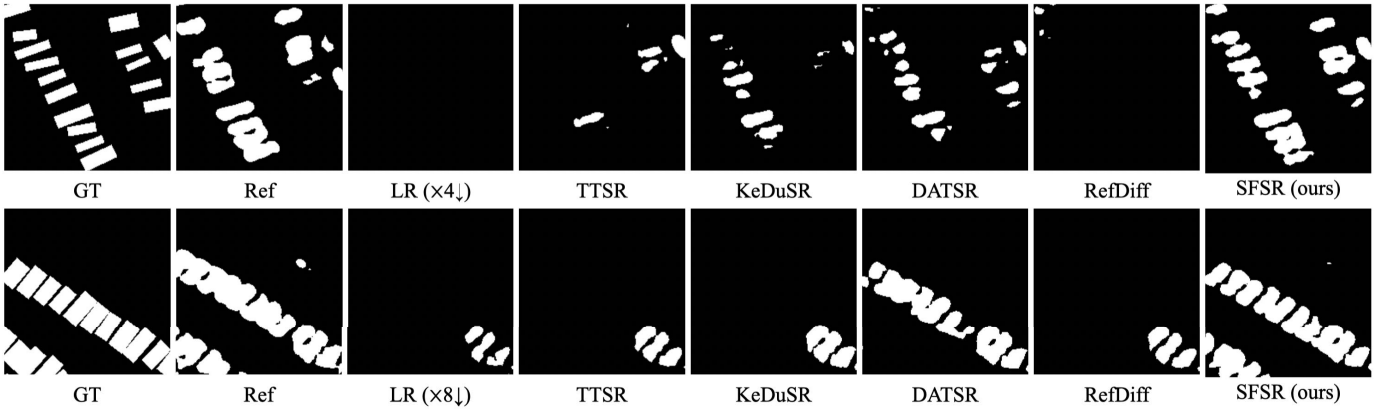


Fig. 11. Qualitative semantic segmentation results for the *small vehicle* class on the UV6K dataset using super-resolved images under synthetically degraded WFOV–NFOV conditions (first row: ×4 WFOV scaling and second row: ×8 WFOV scaling).

TABLE III

DOWNSTREAM DETECTION AND SEGMENTATION PERFORMANCE UNDER SYNTHETIC WFOV–NFOV DEGRADATION AT ×4 AND ×8 SCALES

Task (Dataset, Metric)	Detection (DOTA, $\mathrm{mAP}_{50:95}$ ↑)		Segmentation (UV6K, IoU↑)	
WFOV Scale	×4	×8	×4	×8
Ref	0.060	0.060	0.179	0.179
LR	0.116	0.035	0.032	0.048
LANCZOS	0.149	0.043	0.217	0.083
SwinIR	0.179	0.047	0.141	0.081
SRGAN	0.185	0.053	0.175	0.068
SR3	0.196	0.062	0.118	0.041
EDiffSR	0.173	0.042	0.157	0.068
BSRGAN	0.187	0.075	0.054	0.036
TTSR	0.189	0.060	0.131	0.051
DATSR	0.230	0.196	0.182	0.075
KeDuSR	0.124	0.039	0.157	0.068
RefDiff	0.158	0.091	0.060	0.038
SFSR (ours)	**0.239**	**0.212**	**0.230**	**0.179**

Qualitative examples in Fig. 10 (first row: ×4 and second row: ×8) visually support these quantitative findings. SFSR preserves fine structural cues that are critical for reliable object detection, especially for small or elongated objects. In contrast, TTSR and KeDuSR often fail to recover such structures under NFOV degradation, leading to missed detections for classes such as *harbor*. RefDiff produces visually sharp reconstructions; however, deviations in object shape or spatial layout from the ground truth can still adversely affect detection accuracy.

2) Segmentation Task on UV6K: We evaluate the impact of super-resolution on vehicle segmentation using the UV6K dataset with a pretrained FarSeg model [35]. Segmentation performance is measured using intersection over union (IoU). Vehicle segmentation is particularly sensitive to reconstruction quality, as even small errors in boundary shape or spatial alignment can significantly degrade pixel-level segmentation accuracy.

Under the synthetic WFOV–NFOV degradation setting, SFSR achieves the best segmentation performance among all compared methods. At ×4 WFOV super-resolution, SFSR attains the highest IoU of 0.230, outperforming all RefSR baselines. At ×8, SFSR maintains an IoU of 0.179, remaining close to the reference baseline and substantially higher than most competing methods.

Qualitative results in Fig. 11 further illustrate these trends. Under severe NFOV degradation, TTSR and KeDuSR fail to recover clear object boundaries, resulting in fragmented or incomplete vehicle masks when processed by FarSeg. RefDiff generates visually sharp images, but distortions in object shape and spatial layout often prevent reliable segmentation. In contrast, super-resolved images produced by SFSR preserve boundary clarity and spatial consistency, enabling FarSeg to produce more complete and coherent vehicle segmentation masks.

These results demonstrate that SFSR better preserves segmentation-relevant spatial structure under degraded reference conditions, leading to more reliable downstream segmentation performance.

C. Evaluation on Real Cross-Sensor Data

To evaluate the generalization ability of SFSR under real cross-sensor conditions, we conduct experiments on NAIP (National Agriculture Imagery Program) imagery [53], [54], which provides image pairs covering the same geographic regions but acquired at different times, with different sensors, and at different native spatial resolutions [55], [56]. In our setting, 0.6-m GSD NAIP images [54] are treated as the higher resolution source and downsampled by factors of ×4 and ×8 to generate WFOV (LR) inputs, corresponding to effective GSDs of 2.4 and 4.8 m, while independently acquired 1.0-m GSD NAIP images [53] are used as NFOV reference images.

Compared to the synthetic datasets (UCM-LUD, DOTA, and UV6K), NAIP imagery has a coarser native resolution and is captured under real operational conditions. As a result, downsampling the 0.6-m imagery alone introduces severe information loss, yielding a conservative and realistic WFOV degradation scenario without relying on artificial noise or blur models. In this regime, spatial resolution loss is often more destructive than lens-induced blur or additive noise, leading to lower absolute reconstruction performance across all methods.

Unlike the synthetic WFOV–NFOV setting, which mainly emphasizes controlled geometric misalignment and optical

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TABLE IV

CLASS-WISE OBJECT DETECTION PERFORMANCE ($AP_{50:95}$) OF REFSR METHODS ON THE DOTA DATASET UNDER SYNTHETICALLY DEGRADED WFOV–NFOV CONDITIONS

(a) WFOV $\times 4$ super-resolution.

Scale	$1\times$		**WFOV** $\times 4$											
Class	HR	NFOV	WFOV	LANCZOS	SwinIR	SR3	SRGAN	EDiffSR	BSRGAN	TTSR	DATSR	KeDuSR	RefDiff	SFSR
baseball diamond	0.191	0.052	0.088	0.076	0.105	0.025	0.106	0.099	0.116	0.007	**0.157**	0.061	0.093	0.130
basketball court	0.219	0.046	0.084	0.111	0.118	0.016	0.116	0.075	0.128	0.012	**0.170**	0.038	0.103	0.169
ground track	0.225	0.114	0.123	0.150	0.153	0.040	0.171	0.163	0.176	0.023	0.190	0.118	0.091	**0.202**
roundabout	0.094	0.009	0.026	0.029	0.041	0.008	0.029	0.040	0.021	0.001	0.038	0.023	0.025	**0.043**
soccer field	0.193	0.055	0.066	0.074	0.083	0.012	0.112	0.060	0.136	0.009	**0.127**	0.059	0.073	0.118
storage tank	0.267	0.035	0.093	0.126	0.149	0.091	0.145	0.142	0.134	0.102	0.188	0.119	0.126	**0.204**
swimming pool	0.020	0.003	0.005	0.005	0.007	0.004	0.009	0.005	0.005	0.006	0.012	0.005	0.006	**0.013**
tennis court	0.798	0.267	0.323	0.524	0.502	0.249	0.543	0.540	0.551	0.032	0.699	0.295	0.483	**0.726**
bridge	0.060	0.006	0.027	0.030	0.041	0.017	0.036	**0.057**	0.031	0.009	0.048	0.019	0.050	0.051
harbor	0.400	0.069	0.200	0.233	0.292	0.178	0.296	0.254	0.231	0.077	0.355	0.168	0.259	**0.368**
helicopter	0.113	0.004	0.005	0.009	0.009	0.007	0.008	0.020	0.009	0.003	**0.037**	0.012	0.007	0.025
large vehicle	0.629	0.054	0.217	0.361	0.502	0.320	0.509	0.495	0.538	0.060	0.564	0.469	0.440	**0.584**
plane	0.505	0.048	0.188	0.189	0.257	0.179	0.258	0.242	0.236	0.063	0.328	0.187	0.215	**0.345**
ship	0.447	0.037	0.128	0.175	0.224	0.128	0.237	0.206	0.257	0.055	0.281	0.125	0.210	**0.322**
small vehicle	0.360	0.108	0.152	0.146	0.204	0.137	0.206	0.191	0.228	0.128	0.250	0.163	0.181	**0.281**

(b) WFOV $\times 8$ super-resolution.

Scale	$1\times$		**WFOV** $\times 8$											
Class	HR	NFOV	WFOV	LANCZOS	SwinIR	SR3	SRGAN	EDiffSR	BSRGAN	TTSR	DATSR	KeDuSR	RefDiff	SFSR
baseball diamond	0.191	0.052	0.013	0.040	0.051	0.010	0.064	0.046	0.088	0.006	0.131	0.047	0.051	**0.132**
basketball court	0.219	0.046	0.003	0.003	0.004	0.020	0.007	0.002	0.009	0.008	0.108	0.003	0.057	**0.140**
ground track	0.225	0.114	0.038	0.072	0.066	0.013	0.083	0.065	0.149	0.041	0.165	0.043	0.054	**0.185**
roundabout	0.094	0.009	0.002	0.003	0.003	0.002	0.004	0.003	0.008	0.006	0.029	0.003	0.015	**0.041**
soccer field	0.193	0.055	0.018	0.028	0.028	0.012	0.039	0.026	0.053	0.017	0.089	0.021	0.040	**0.121**
storage tank	0.267	0.035	0.032	0.050	0.051	0.032	0.048	0.048	0.084	0.054	0.159	0.050	0.077	**0.165**
swimming pool	0.020	0.003	0.002	0.003	0.003	0.001	0.003	0.004	0.004	0.004	0.009	0.003	0.003	**0.011**
tennis court	0.798	0.267	0.057	0.083	0.088	0.027	0.106	0.072	0.152	0.051	0.634	0.038	0.277	**0.661**
bridge	0.060	0.006	0.012	0.012	0.012	0.012	0.013	0.012	0.014	0.008	**0.040**	0.010	0.033	0.039
harbor	0.400	0.069	0.070	0.065	0.073	0.079	0.078	0.068	0.104	0.086	0.312	0.060	0.142	**0.332**
helicopter	0.113	0.004	0.003	0.002	0.003	0.007	0.003	0.005	0.002	0.004	**0.049**	0.002	0.009	0.015
large vehicle	0.629	0.054	0.038	0.070	0.077	0.066	0.104	0.057	0.137	0.143	0.481	0.080	0.259	**0.521**
plane	0.505	0.048	0.071	0.057	0.072	0.085	0.069	0.058	0.103	0.072	0.279	0.070	0.126	**0.309**
ship	0.447	0.037	0.044	0.041	0.046	0.045	0.051	0.044	0.056	0.066	0.232	0.043	0.120	**0.269**
small vehicle	0.360	0.108	0.117	0.114	0.121	0.123	0.120	0.114	0.161	0.130	0.222	0.117	0.100	**0.242**

degradation, NAIP imagery is accurately georeferenced and thus exhibits minimal spatial misalignment. However, it introduces a more challenging and realistic form of reference unreliability arising from cross-sensor and cross-temporal inconsistencies. Specifically, WFOV inputs (downsampled from 0.6-m imagery) and NFOV references (1.0-m imagery) are typically captured by different sensors and approximately ten years apart, resulting in natural variations in radiometric characteristics, land-cover patterns, and man-made structures. This setting makes direct reference utilization inherently unreliable even when geometric alignment is accurate, providing a realistic real-world evaluation scenario that complements the synthetic experiments.

Quantitative results are summarized in Table V. Notably, most RefSR methods perform worse than SISR baselines in this setting. This behavior arises because temporally inconsistent NFOV references often contain scene structures that no longer exist in the WFOV input. When such references are directly fused, they can mislead the reconstruction process and degrade image quality instead of providing useful guidance. In contrast, SFSR achieves the highest SSIM at both $\times 4$ and $\times 8$ WFOV scaling factors while maintaining competitive perceptual quality in terms of LPIPS and FID. These results indicate that SFSR better preserves structural consistency under cross-temporal reference mismatch.

TABLE V

QUANTITATIVE RESULTS ON THE REAL MULTISENSOR REMOTE SENSING DATASET (NAIP) [53], [54], COMPARING SFSR WITH STATE-OF-THE-ART METHODS

WFOV scale	$\times 4$		$\times 8$	
Metric	PSNR↑/SSIM↑	LPIPS↓/FID↓	PSNR↑/SSIM↑	LPIPS↓/FID↓
Ref	16.71 / 0.448	0.239 / 35.20	16.71 / 0.448	0.239 / 35.20
LR	21.45 / 0.423	0.146 / 28.78	19.81 / 0.323	0.289 / 149.47
LANCZOS	23.64 / 0.486	0.045 / 45.84	23.05 / 0.415	0.270 / 108.47
SwinIR	**25.66** / 0.605	0.009 / 33.24	**23.82** / 0.448	0.169 / 97.67
SRGAN	25.03 / 0.589	0.009 / 36.08	23.81 / 0.442	0.150 / 90.32
SR3	23.46 / 0.498	0.007 / 21.46	22.08 / 0.316	0.091 / 51.47
EDiffSR	24.69 / 0.591	**0.007** / **20.91**	23.79 / 0.417	0.101 / 44.04
BSRGAN	23.40 / 0.495	0.037 / 36.59	22.20 / 0.422	0.097 / 43.99
TTSR	23.16 / 0.537	0.039 / 36.24	23.07 / 0.421	0.175 / 105.28
DATSR	23.66 / 0.538	0.037 / 35.88	23.04 / 0.419	0.218 / 96.08
KeDuSR	23.27 / 0.536	0.045 / 34.64	23.20 / 0.442	0.144 / 83.50
RefDiff	21.11 / 0.447	0.033 / 32.19	20.98 / 0.401	**0.041** / 38.95
SFSR (ours)	23.77 / **0.609**	0.014 / 21.37	22.41 / **0.474**	0.073 / **37.52**

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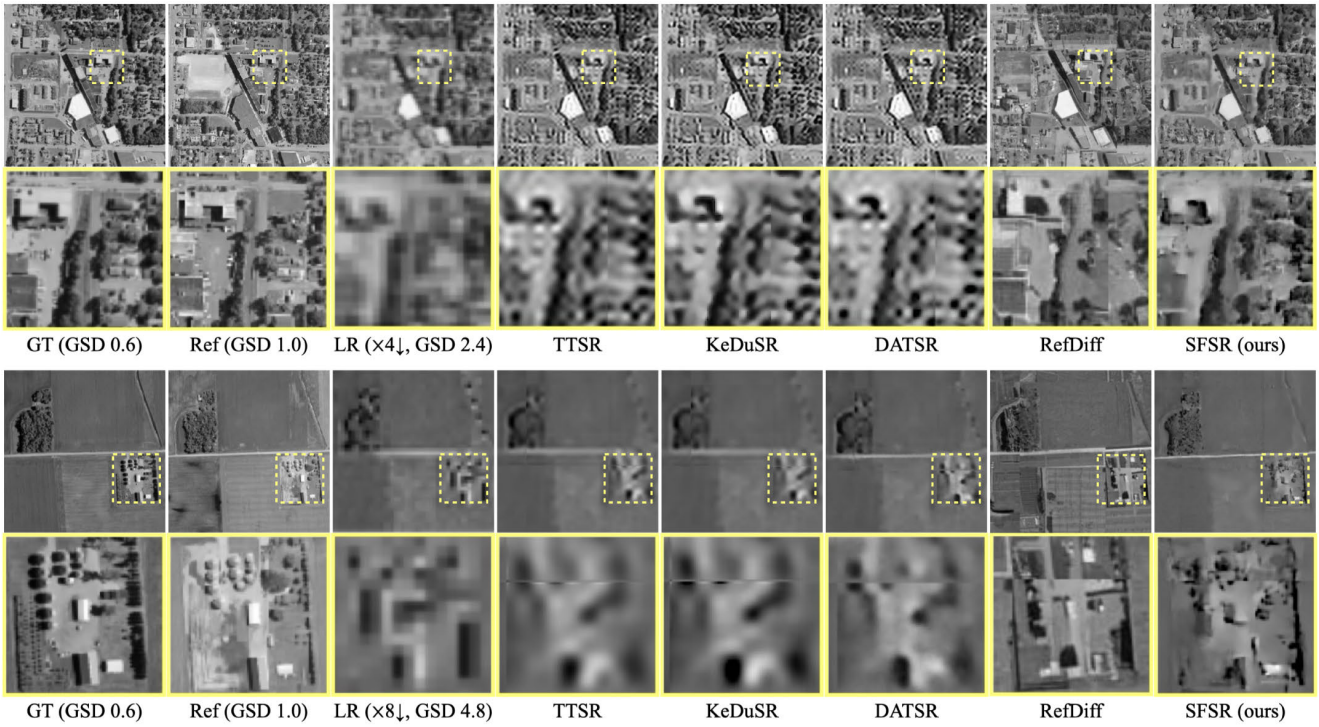


Fig. 12. Qualitative comparison of RefSR methods on the real cross-sensor NAIP dataset (first row: ×4 WFOV super-resolution and second row: ×8).

TABLE VI

ABLATION STUDY ON THE UCM-LUD DATASET UNDER SYNTHETIC WFOV–NFOV DEGRADATION, ANALYZING THE EFFECTS OF ATTENTION MECHANISMS, SHARED ENCODING, AND FREQUENCY-AWARE LOSS COMPONENTS AT ×4 AND ×8 WFOV SCALES

Variations	Attention Methods		Spectral Fusion			WFOV scale ×4		WFOV scale ×8	
	Cross-Attn	SWCA	Shared Encoder	LF loss	NM loss	PSNR↑ / SSIM↑	LPIPS↓ / FID↓	PSNR↑ / SSIM↑	LPIPS↓ / FID↓
A	✓					25.37 / 0.717	0.194 / 163.22	22.79 / 0.635	0.272 / 190.09
B	✓		✓			25.50 / 0.726	0.193 / 161.92	23.38 / 0.656	0.254 / 177.35
C	✓		✓	✓		26.25 / 0.744	0.194 / 157.16	24.86 / 0.690	0.227 / 170.92
D	✓		✓	✓	✓	26.19 / 0.751	0.193 / 156.38	24.61 / 0.705	0.220 / 166.49
E		✓				25.95 / 0.763	0.191 / 155.58	24.01 / 0.701	0.222 / 167.64
F		✓	✓			26.57 / 0.779	0.192 / 154.11	24.67 / 0.718	0.217 / 168.22
G		✓	✓	✓		27.53 / 0.806	0.190 / 152.76	25.31 / 0.757	0.200 / 163.01
H		✓	✓	✓	✓	**27.81 / 0.813**	**0.189 / 150.55**	**25.83 / 0.762**	**0.199 / 162.90**

Qualitative comparisons are shown in Fig. 12, where only RefSR methods are presented for clarity. TTSR, DATSR, and KeDuSR struggle to reliably recover fine structures from temporally inconsistent NFOV references, often producing blurred or incomplete details. RefDiff generates visually sharp images but frequently introduces scene elements that deviate from the actual geometry. In contrast, SFSR produces more conservative reconstructions that preserve existing scene structures, selectively exploiting only reliable reference cues while avoiding large geometric deviations. This behavior highlights the robustness of SFSR in handling real cross-sensor and cross-temporal inconsistencies without amplifying misleading reference information.

D. Ablation and Efficiency Analysis

In this section, we analyze the internal behavior and practical efficiency of SFSR through ablation studies, reproducibility and stability analysis, and efficiency evaluation. All experiments in this section are conducted on the UCM-LUD dataset under the synthetic NFOV–WFOV degradation setting, using ×4 and ×8 WFOV super-resolution.

1) Ablation Study: The ablation results are summarized in Table VI. Comparing variants A–D with E–H shows that replacing standard cross-attention with the SWCA mechanism consistently improves performance across all metrics and WFOV scaling factors. This gain is particularly noticeable under degraded NFOV reference conditions, where localized window-based attention reduces the influence of spatial misalignment and unreliable reference responses.

Introducing the shared encoder further improves both pixel-level accuracy and perceptual quality. This suggests that embedding WFOV and NFOV inputs into a common feature space enables more effective and coherent reference interaction. The low-frequency alignment (LF) loss and noise minimization (NM) loss provide additional performance gains, especially at larger upscaling factors, where reference degradation and noise effects become more severe.

Overall, each component contributes positively and complements the others. The full SFSR configuration achieves

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TABLE VII

REPRODUCIBILITY AND STABILITY OF LEARNED SPECTRAL BOUNDARIES AND RECONSTRUCTION PERFORMANCE ACROSS REPEATED RUNS AND DATA SPLITS (MEAN WITH STANDARD DEVIATION IN PARENTHESES)

WFOV scale	×4		×8	
Eval Protocol	τ_L / τ_H	PSNR / SSIM	τ_L / τ_H	PSNR / SSIM
5× Runs	0.38 / 19.0 (0.01) / (0.0)	27.81 / 0.813 (0.32) / (0.011)	0.19 / 19.0 (0.03) / (0.0)	25.83 / 0.762 (0.29) / (0.013)
5× Splits	0.43 / 19.0 (0.02) / (0.0)	25.30 / 0.759 (0.94) / (0.022)	0.26 / 19.0 (0.06) / (0.0)	23.15 / 0.687 (1.14) / (0.035)

TABLE VIII

PARAMETER SIZE AND GPU MEMORY USAGE OF DIFFUSION-BASED SR MODELS (BATCH SIZE: 4)

Models	SR3 [1]	EDiffSR [47]	RefDiff [38]	SFSR (ours)
parameter size	37.94 M	26.79 M	116.74 M	69.96 M
memory footprint (training, 4 batch)	1,526 MB	24,218 MB	15,603 MB	3,670 MB
memory footprint (inference, 4 batch)	716 MB	1,840 MB	3,058 MB	1,194 MB

the best and most stable performance across all settings, confirming the effectiveness of the proposed frequency-aware reference utilization design.

2) Reproducibility and Stability: Table VII reports reproducibility results from five repeated runs and five different data splits on the UCM-LUD dataset under the synthetic NFOV–WFOV degradation setting. For PSNR and SSIM, we report the average of the best-performing checkpoints from each run or split, while the spectral boundaries τ_L and τ_H are averaged over the corresponding best-performing checkpoints.

For both ×4 and ×8 WFOV super-resolution, the high-frequency boundary τ_H, which defines the band used to isolate noise-dominated high-frequency responses, consistently converges to the highest frequency index with negligible variation across runs. The larger variation of τ_L reflects its role in adaptively adjusting the balance between low- and MF reference features depending on the reconstruction difficulty.

Reconstruction performance is similarly stable. PSNR and SSIM vary little across repeated runs for both scales. Under different data splits, absolute performance decreases for both ×4 and ×8, while performance variance becomes more pronounced at ×8, where the reconstruction task is more challenging.

3) Efficiency Analysis: We evaluate the efficiency of SFSR in terms of sampling cost, inference time, and memory usage. When the number of reverse diffusion steps is reduced from 200 to 20, only a modest degradation in reconstruction quality is observed, as summarized in Tables I and II. Despite this reduction, both structural and perceptual performances remain largely preserved.

In contrast, inference time is significantly reduced from 3.08 to 0.39 s, corresponding to nearly an eightfold speedup. This demonstrates that SFSR can achieve high-quality reconstruction with substantially fewer sampling steps, making it practical for large-scale or time-sensitive remote sensing applications.

Table VIII compares GPU memory usage among diffusion-based super-resolution methods. SFSR requires considerably less memory than RefDiff and EDiffSR during both training and inference, benefiting from its structured and frequency-aware reference utilization. Overall, SFSR achieves a favorable balance between reconstruction quality and computational efficiency.

VI. CONCLUSION AND DISCUSSION

This study presents SFSR, a diffusion-based reference super-resolution framework designed for realistic remote sensing scenarios in which NFOV reference images are degraded by noise, blur, and spatial inconsistency. Unlike conventional RefSR methods that assume clean or well-aligned references, SFSR introduces frequency-aware reference utilization to robustly integrate imperfect NFOV references with WFOV inputs.

Extensive experiments demonstrate that SFSR consistently outperforms existing RefSR methods in terms of reconstruction accuracy, perceptual quality, and downstream tasks such as object detection and semantic segmentation. These results indicate that controlled spectral fusion and selective reference utilization are more reliable than direct texture transfer or alignment-dependent strategies, particularly when reference reliability cannot be guaranteed. While SFSR also achieves competitive performance compared to SISR methods, its primary contribution lies in advancing RefSR under realistic multisensor degradation conditions rather than replacing SISR paradigms.

Experiments on the NAIP dataset [53], [54] further validate the robustness of SFSR under real cross-sensor settings involving resolution mismatch and temporal inconsistency. Although NAIP does not represent true WFOV–NFOV image pairs captured by a single multisensor satellite platform with distinct optics, it introduces practical challenges arising from cross-resolution and cross-temporal discrepancies. Under these conditions, SFSR maintains stronger structural consistency than existing RefSR methods, highlighting its ability to avoid misleading reference cues when reference information is unreliable.

Our analysis also highlights the importance of the edge mask threshold τ_{mask}, which governs how high-frequency reference responses are treated as noise or structural detail. As discussed in Section III, τ_{mask} plays a central role in controlling frequency-aware reference utilization and directly influences the redistribution of reference information across frequency bands. While a fixed value yields stable behavior across our experiments, τ_{mask} remains a data-dependent parameter, and its automatic or adaptive selection represents an important direction for future work.

Looking forward, an important avenue for future research is the evaluation of SFSR on authentic WFOV and NFOV imagery acquired from operational multicamera satellite systems with distinct optics and acquisition geometries. Such data would enable more rigorous assessment of reference degradation, spatial inconsistency, and spectral mismatch under physically characterized sensor relationships. In addition, while this work focuses on RefSR with unreliable auxiliary

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views, the proposed frequency-aware feature redistribution mechanism may be extended to sensor-fusion super-resolution frameworks, such as pan-sharpening or MTF-aware SR, where regulating frequency transfer in a physically meaningful manner is critical. Exploring these connections remains an interesting direction for future study.

In summary, SFSR establishes a strong and practical baseline for RefSR in degraded remote sensing environments. By explicitly regulating reference utilization in the frequency domain, the proposed framework demonstrates that frequency-aware fusion is a principled and effective strategy for robust RefSR, providing a solid foundation for future research on multisensor satellite super-resolution using heterogeneous WFOV–NFOV observations.

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